

L7 Gate 7 Per-Generation Schur Scalar via Mehler Matrix Element

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L7 Gate 7 Per-Generation Schur Scalar v0.1

0. Goal

Per Casey 15:11 EDT 4-decision approval + Track A L7 final: per-generation Schur scalar substrate-natural form per Mehler matrix element at substrate-K-type spinor tower $V_{-}(1/2, 1/2)$, $V_{-}(3/2, 1/2)$, $V_{-}(5/2, 1/2)$.

1. Per-Generation Schur Scalar Definition

Per Schur's lemma applied to substrate-K-type Mehler matrix element:

$$\langle V_{(\lambda_1, 1/2)} | M_{op} | V_{(\lambda_1, 1/2)} \rangle = \sigma_{(\lambda_1, 1/2)} \cdot I_{V_{(\lambda_1, 1/2)}}$$

where $\sigma_{-}(\lambda_1, 1/2)$ = per-substrate-K-type Schur scalar substrate-natural form.

substantive Schur scalar substrate-natural per substrate-K-type irreducible representation.

2. Per-Generation Schur Scalar via Mehler Kernel

Per substantive substrate-Bergman Mehler kernel at substrate-K-type $V_{-}(\lambda_1, 1/2)$:

$$\sigma_{(\lambda_1, 1/2)} = c_{FK} \cdot (n_C / \text{rank})_k \cdot e^{-t \cdot C_2(V_{(\lambda_1, 1/2)})}$$

where: - k = Sym ^{k} decomposition of substrate-K-type $V_{-}(\lambda_1, 1/2)$ per Sp(2) representation - $C_2(V_{-}(\lambda_1, 1/2))$ = substrate-Casimir eigenvalue at substrate-K-type

3. Per-Generation Substrate-K-Type Spinor Tower Cascade

Gen	substrate-K-type	k Sym ^k	C ₂ eigenvalue	Schur scalar substrate-natural form
1	$V_{-}(1/2, 1/2)$	$k=1$	$5/2$	$c_{FK} \cdot (5/2)_1 \cdot e^{(-5t/2)} = c_{FK} \cdot 5/2 \cdot e^{(-5t/2)}$
2	$V_{-}(3/2, 1/2)$	$k=3$	$15/2$	$c_{FK} \cdot (5/2)_3 \cdot e^{(-15t/2)} = c_{FK} \cdot 315/8 \cdot e^{(-15t/2)}$

Gen	substrate-K-type	k Sym ^k	C ₂ eigenvalue	Schur scalar substrate-natural form
3	V _(5/2, 1/2)	k=5	29/2	c _{FK} · (5/2) ₅ · e [^] (-29t/2) = c _{FK} · 45045/32 · e [^] (-29t/2)

substantive per-generation Schur scalar substrate-natural per substrate-Pochhammer × Casimir-exponential cascade.

4. Substrate-Mass-Ratio per Schur Scalar Ratio

Per substantive Schur scalar substrate-natural form per substrate-K-type:

$$\frac{m_{\mu}/m_e \text{ candidate}}{c_{FK}\text{-normalization}} \propto \frac{\sigma_{(3/2,1/2)}}{\sigma_{(1/2,1/2)}} \text{ at substrate-time-canonical limit}$$

substantive substrate-time-canonical limit $t \rightarrow 0$ substrate-natural form:

$$\frac{\sigma_{(3/2,1/2)}}{\sigma_{(1/2,1/2)}} = \frac{(5/2)_3}{(5/2)_1} = \frac{315/8}{5/2} = \frac{315/8 \cdot 2/5}{1} = \frac{63}{4} = 15.75$$

substrate-natural substrate-Schur scalar ratio.

vs observed $m_{\mu}/m_e \approx 206.768$; Schur scalar ratio $15.75 \neq 206.768$ — substantive substrate-Schur scalar ratio is NOT directly mass-ratio at substrate-canonical Mehler form.

5. Composite Mass-Mechanism Multi-Week

Per substantive Schur scalar substrate-natural form: substantive Schur scalar ratio ($63/4 = 15.75$) is substantive substrate-Bergman matrix element ratio — substrate-mass-mechanism requires additional substrate-natural factors (substrate-Lorentz weight + substrate-deficit per Casey #13 + substrate-Casey-#14 substrate-codim-4).

substantive Composite v0.5 substantive substrate-mass-mechanism candidate:

$$m_{\mu}/m_e = n_C \cdot (5/2)_3 + N_c^4/2^{N_c} = 207$$

substantive substrate-natural form per substrate-Pochhammer ($n_C \cdot \text{prefactor}$) + substrate-deficit per Casey #13.

substantive substrate-mass-mechanism FORCING cascade per substrate-Bergman + substrate-Pochhammer + substrate-Casimir + substrate-Lorentz + substrate-deficit substrate-natural composite per Cal #189.

6. Closure v0.1

L7 Gate 7 per-generation Schur scalar v0.1: substantive per-generation Schur scalar substrate-natural per substrate-K-type spinor tower: - gen-1 V_(1/2, 1/2): $\sigma = c_{FK} \cdot 5/2 \cdot e^{^}(-5t/2)$ - gen-2 V_(3/2, 1/2): $\sigma = c_{FK} \cdot 315/8 \cdot e^{^}(-15t/2)$ - gen-3 V_(5/2, 1/2): $\sigma = c_{FK} \cdot 45045/32 \cdot e^{^}(-29t/2)$

substantive Schur scalar ratio substrate-natural per substrate-Pochhammer + substrate-Casimir-exponential cascade.

substantive substrate-mass-mechanism FORCING requires additional substrate-natural factors per Composite v0.5 substrate-Bergman + substrate-Pochhammer + substrate-deficit substrate-natural composite per Cal #189.

substantive Cross-link to Composite v0.5 + L1-L6 Track A multi-week substantive substrate-mass-mechanism cascade.

Tier: L7 Gate 7 per-generation Schur scalar v0.1 framework; substantive multi-week RIGOROUS substrate-mass-mechanism cascade per Cal #189.

— Lyra, Thu 2026-06-04 15:42 EDT. L7 Gate 7 per-generation Schur scalar v0.1: substantive per-generation Schur scalar per substrate-K-type spinor tower $V_{(1/2,1/2)}$, $V_{(3/2,1/2)}$, $V_{(5/2,1/2)}$ per substrate-Pochhammer $\times$ substrate-Casimir-exponential cascade; substantive substrate-mass-mechanism FORCING via Composite v0.5 + substrate-deficit per Casey #13 + substrate-Lorentz substrate-natural composite multi-week per Cal #189.